identical spins and thus provide the detectors they enter with identical programs. Thus, if the same axes are chosen for the left and right detectors, the spin detectors will find identical results.

Notice that these spin detectors exactly reproduce everything encountered by Scully and Mulder, though with simple substitutions: instead of choosing a door on a titanium box, we are choosing an axis; instead of seeing a red or blue flash, we record a clockwise or counterclockwise spin. So, just as opening the same doors on a pair of identically numbered titanium boxes results in the same color flashing, choosing the same axes on the two detectors results in the same spin direction being measured. Also, just as opening one particular door on a titanium box prevents us from ever knowing what color would have flashed had we chosen another door, measuring the electron spin about one particular axis prevents us, via quantum uncertainty, from ever knowing which spin direction we would have found had we chosen a different axis.

All of the foregoing means that Mulder's analysis of how to learn who's right applies in exactly the same way to this situation as it does to the case of the alien spheres. If EPR are correct and each electron actually has a definite spin value about all three axes—if each electron provides a "program" that definitively determines the result of any of the three possible spin measurements—then we can make the following prediction. Scrutiny of data gathered from many runs of the experiment—runs in which the axis for each detector is randomly and independently selected—will show that *more than half the time, the two electron spins agree, being both clockwise or both counterclockwise.* If the electron spins do not agree more than half the time, then Einstein, Podolsky, and Rosen are wrong.

This is Bell's discovery. It shows that even though you can't actually measure the spin of an electron about more than one axis—even though you can't explicitly "read" the program it is purported to supply to the detector it enters—this does *not* mean that trying to learn whether it nonetheless has a definite amount of spin about more than one axis is tantamount to counting angels on the head of a pin. Far from it. Bell found that there is a bona fide, testable consequence associated with a